Magneto2D mesh convergence

Fixed finer FEMM references | normal, fine and very-fine native meshes | identical 25-position cogging grids

All six additional native jobs are complete. Each request differs from its original registered request only in mesh density. Corner refinement, materials, current settings, angular samples and weighted-stress torque remain fixed.

Fixture	Native mesh	Elements	P-p (Nm)	H1 (Nm)	P-p diff	H1 diff	Gates
SPM 8p/24s distributed	normal	32,614	26.344	4.749	10.73%	10.17%	YELLOW
SPM 8p/24s distributed	fine	34,722	25.938	4.666	9.02%	8.25%	PASS
SPM 8p/24s distributed	very fine	105,682	24.524	4.400	3.08%	2.09%	PASS
SPM 8p/48s distributed	normal	44,718	17.827	6.712	9.10%	7.30%	PASS
SPM 8p/48s distributed	fine	41,662	18.405	6.947	12.63%	11.05%	YELLOW
SPM 8p/48s distributed	very fine	126,662	16.875	6.413	3.27%	2.51%	PASS
SPM 12p/36s distributed	normal	39,280	27.010	6.617	11.20%	9.24%	YELLOW
SPM 12p/36s distributed	fine	41,988	26.792	6.508	10.30%	7.45%	YELLOW
SPM 12p/36s distributed	very fine	118,966	25.279	6.246	4.07%	3.12%	PASS

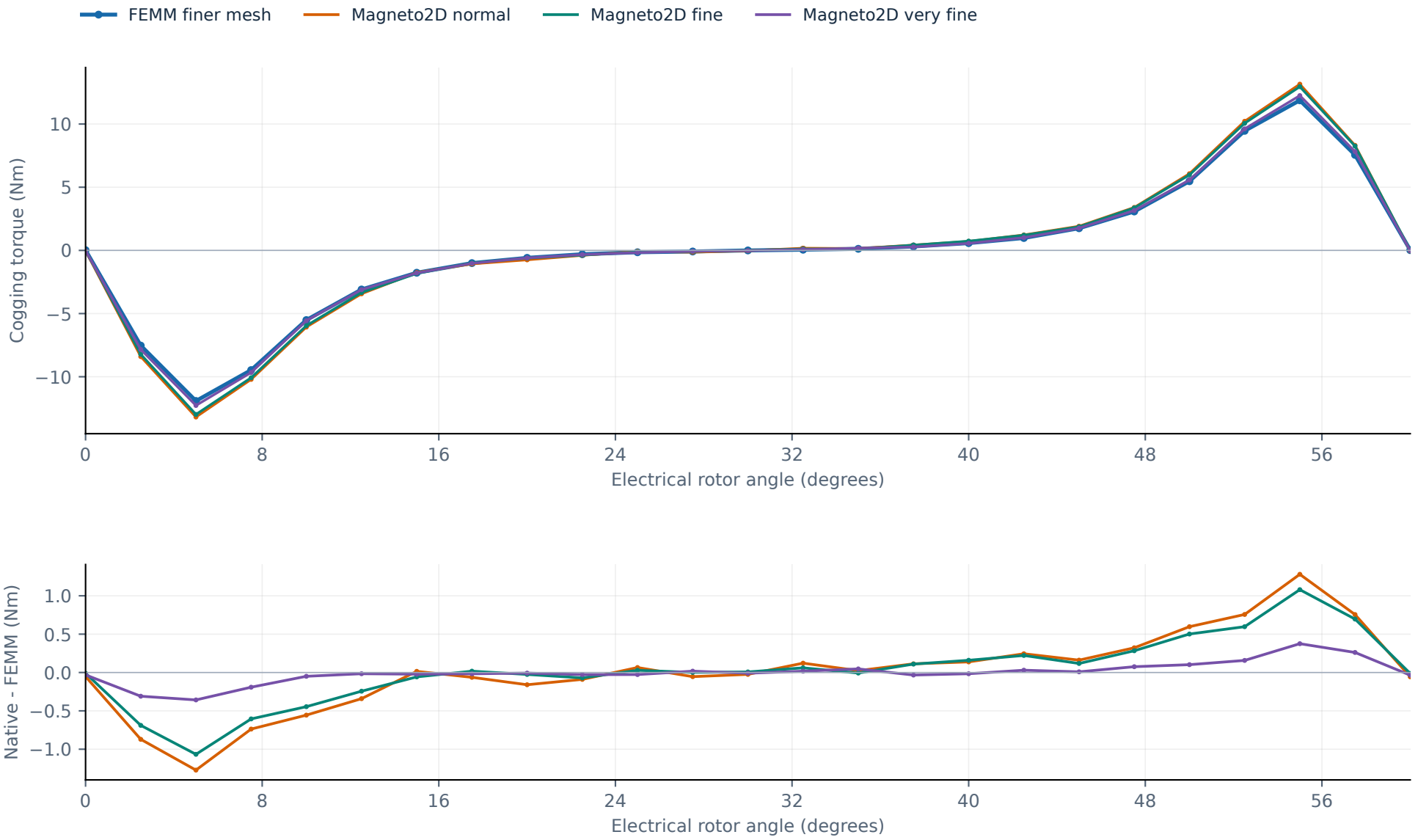
Reading the table. H1 is the fundamental cogging harmonic. Differences are absolute relative to the same finer FEMM reference for every native mesh level. PASS allows max(10%, the fixed case floor); YELLOW allows max(20%, twice that floor). Element counts are representative counts from native solve metadata, rather than all cogging positions.

Scope. The fine preset is a small target-size change and does not guarantee a larger reported element count for every case. Very fine supplies a stronger refinement. This is a diagnostic study, not a change to the registered fixtures, reference bundle or release qualification. Waveforms are compared without fitting, filtering, alignment or rescaling.

NATIVE MESH CONVERGENCE

SPM 8p/24s distributed

All native mesh levels compared with the same completed finer FEMM waveform.



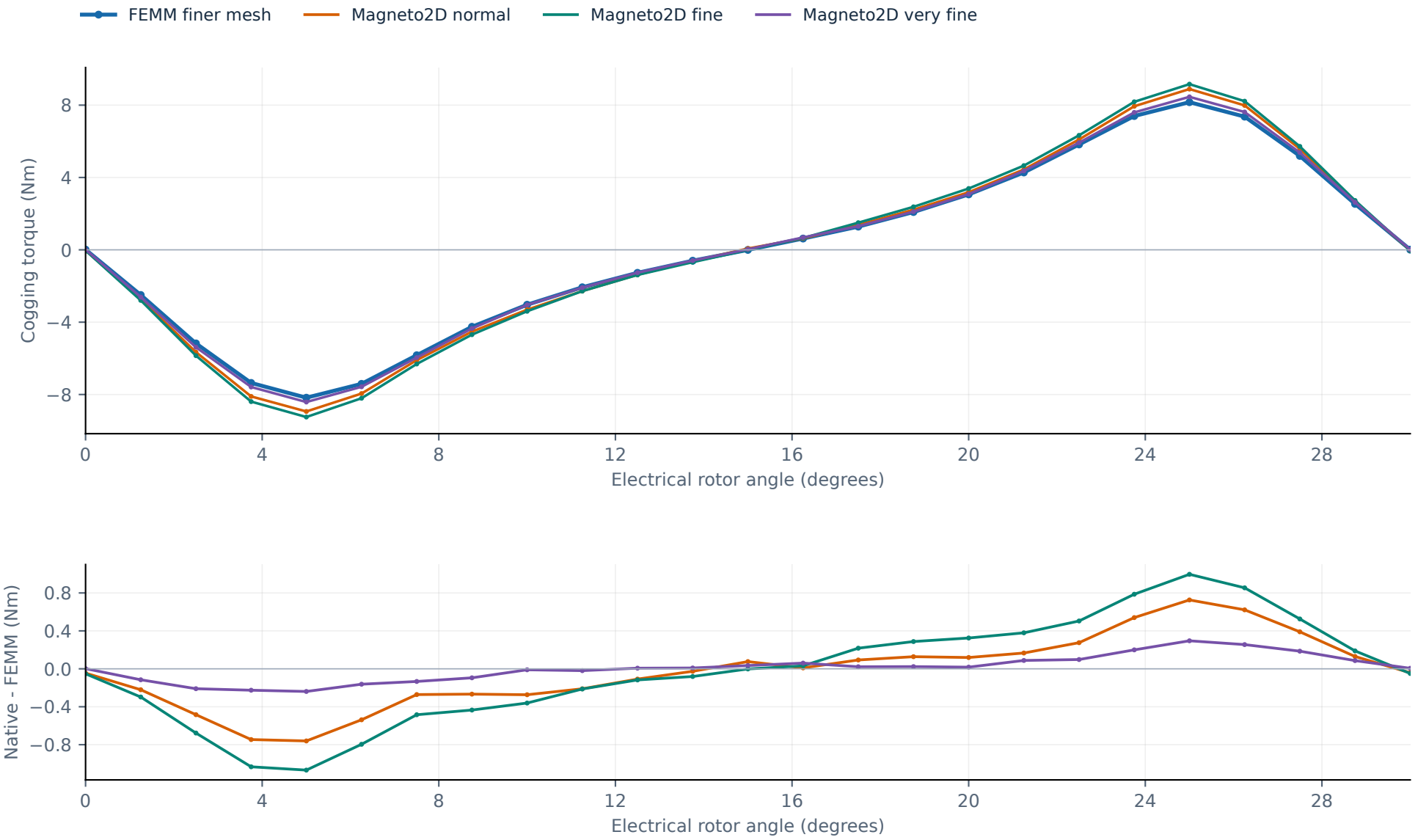
normal: p-p 10.73%, H1 10.17% (YELLOW) | fine: p-p 9.02%, H1 8.25% (PASS)

very fine: p-p 3.08%, H1 2.09% (PASS)

NATIVE MESH CONVERGENCE

SPM 8p/48s distributed

All native mesh levels compared with the same completed finer FEMM waveform.



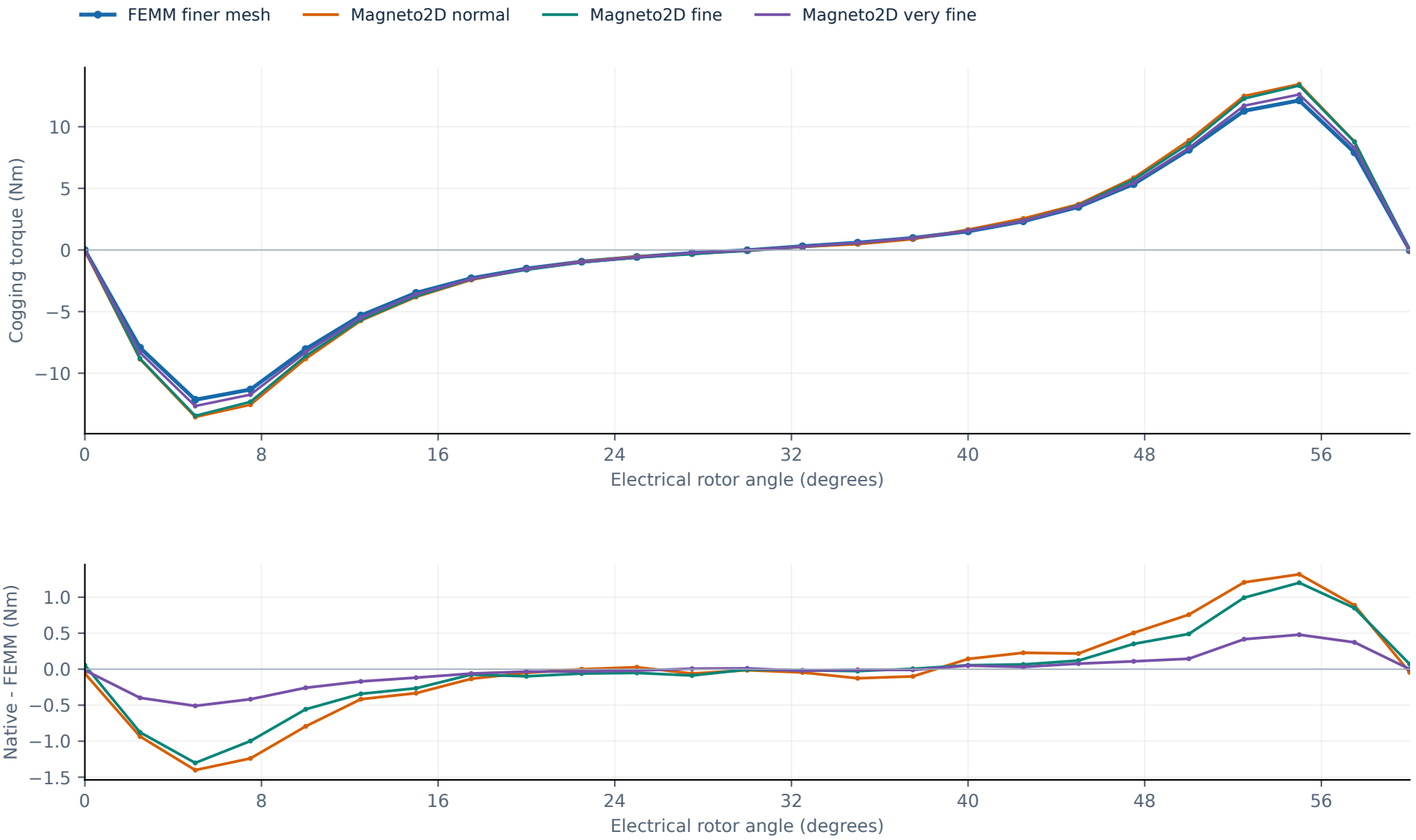
normal: p-p 9.10%, H1 7.30% (PASS) | fine: p-p 12.63%, H1 11.05% (YELLOW)

very fine: p-p 3.27%, H1 2.51% (PASS)

NATIVE MESH CONVERGENCE

SPM 12p/36s distributed

All native mesh levels compared with the same completed finer FEMM waveform.



normal: p-p 11.20%, H1 9.24% (YELLOW) | fine: p-p 10.30%, H1 7.45% (YELLOW)

very fine: p-p 4.07%, H1 3.12% (PASS)